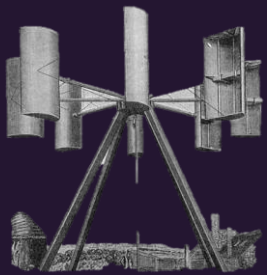




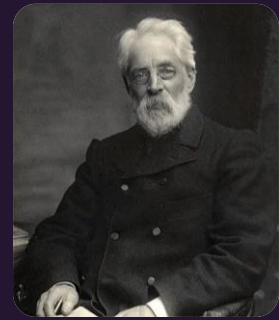
Vertical Axis Wind Turbine:

A compact, urban-ready wind energy prototype that converts variable wind into usable electricity. This project demonstrates a practical, low-maintenance solution for decentralised power.

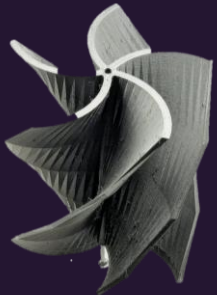
1 . BACKGROUND OF VAWT



1887 — **James Blyth** (Scotland): Built the world's first electricity-generating VAWT to power his cottage.



1922 — **Sigurd Johannes** (Finland): Invented the drag-based S-shaped Savonius turbine, which spins easily in low-speed wind.



1931 — **Georges Jean Marie** (France): Patented the lift-based Darrieus ("eggbeater") turbine, establishing the aerofoil design used in modern high-efficiency VAWTs.



The Highway Divider Concept & Real-World Testing

- 2010–2012 (Scientific Proof): University researchers in India and the US published studies proving that cars moving in opposite directions create a continuous, concentrated wind vortex over median dividers.
- 2017 (First Successful Commercial Trial): Kadir Yürüten (Devecitech) successfully deployed smart ENLIL turbines on Istanbul's highway dividers to power local sensors and streetlights.



COMMUNITY PROBLEM



Electricity consumption exceed production on global stage and we are not capable to utilize it efficiently.

Rising household consumption increases costs and carbon emissions.



The Solution: Harnessing the Power of VAWT

Vertical Axis Wind Turbines capture wind from any direction, making them ideal for turbulent, variable urban flows. Their compact footprint, simplified mounting and reduced maintenance make VAWTs well suited for residential and community-scale deployments.

Omnidirectional

mechanism — captures wind from any direction.

Compact

Smaller form factor fits rooftops and small plots.

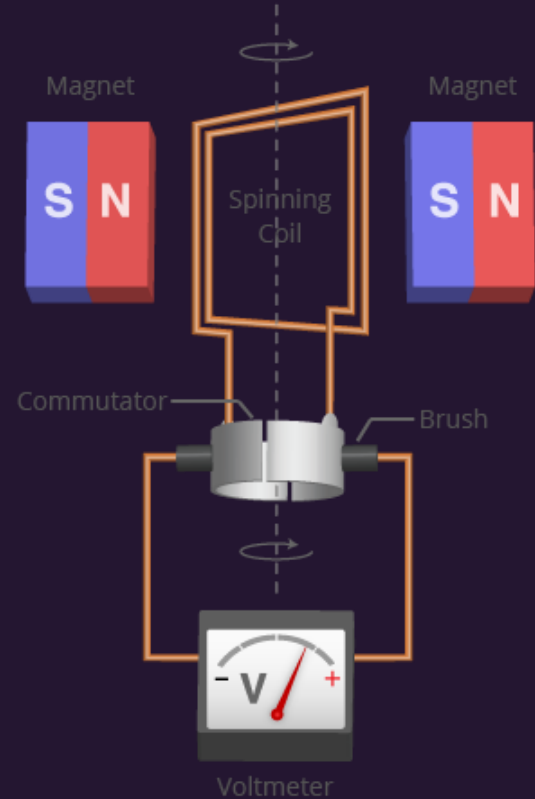
Low Maintenance

Few moving parts and simpler bearings reduce upkeep.

HOW DYNAMO WORKS

A dynamo converts mechanical energy into electrical energy using **electromagnetic induction**.

As a mechanical force rotates a coil of copper wire within a stationary magnetic field, the movement forces electrons in the wire to flow, creating an electric current.



Project Components: The Anatomy of the Turbine



Arduino Uno

Central controller for data acquisition, signal processing and control logic.



Current Sensor

Monitors output current from the dynamo for power calculations and safety.



Voltage Sensor

Measures generated voltage by dynamo



Hall Effect Sensor

Detects RPM by reading magnets on the rotating shaft — essential for performance metrics.



DHT11 Sensor

L temperature and humidity to correlate environmental conditions with output.



Breadboard

Prototyping platform for sensor connections and temporary circuits.



LED Indicators

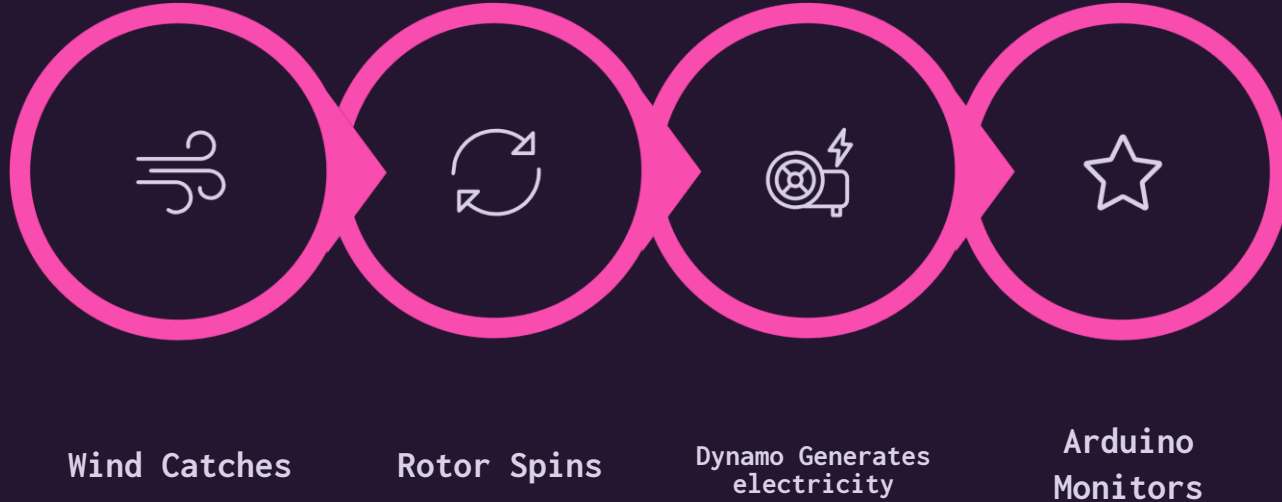
Visual status signals
(.streetlight)



Dynamo Motor

Primary generator converting rotational motion into electrical energy; wired through rectification and regulation stages.

How It Works: The Technical Process



Airflow produces torque on the blades; the rotor spins the dynamo producing alternating current. Power is rectified and conditioned; Arduino reads voltage, current and RPM and logs data for management or storage.

Results: Powering Small-Scale Needs

Prototype testing converted wind into measurable electrical output, demonstrating stable operation across variable wind speeds. Readings showed dependable voltage and current traceable to RPM and environmental conditions — validating the monitoring architecture.

Proof

Proof of Concept

Operational prototype

Scalable

Scalable

Modular design suitable for small arrays or rooftop deployments.

Advantages for a Greener Future



Decentralised Power

Reduces dependence on distant generation; lowers transmission losses.



Lower Emissions

Clean electricity locally reduces carbon footprint for homes and communities.



Quiet & Safe

Reduced noise and safer operation compared with large horizontal-axis turbines.

Conclusion: The Winds of Change

This VAWT project illustrates a practical path to localised, sustainable electricity. With reliable sensing and simple control, small turbines can support off-grid power and reduce pressure on central grids.



THANK YOU

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